AYŞE FEYZA BİRER

Electronics Engineering Student



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PROFESSIONAL SUMMARY

Fourth-year Electronics Engineering student at Gebze Technical University focused on analog/RF and digital integrated circuit design. Hands-on experience with the full-custom IC design flow in 90 nm CMOS using Synopsys Custom Compiler, HSPICE and StarRC, from transistor sizing and schematic design to layout, DRC/LVS and post-layout verification, as well as analog circuit simulation in LTspice. Designed a 2.4 GHz CMOS low-noise amplifier that met all performance targets within its power budget, and developed Python-based simulation and algorithm work as a member of the Kıray Air Defense TEKNOFEST team. Completed a long-term internship in the Avionics Systems Development unit at Baykar and is currently working on a graduation project on machine learning–based aging prediction and self-healing analog circuits. Interested in VLSI design, analog/mixed-signal circuits and embedded systems.

EXPERIENCE

Baykar — Long-Term Intern **Oct 2026 - Jan 2027**

Avionics Systems Development Unit | Türkiye

- Completed a long-term internship in the Avionics Systems Development unit, contributing to development activities for avionics systems.

Kıray Air Defense — Software Team Member **2024 - 2025**

TEKNOFEST Project | Gebze, Türkiye

- Carried out software design, algorithm development and simulation studies in Python for a project focused on communication systems.
- Developed and tested system behavior in PyBullet simulation environments before hardware testing.
- Took an active role in hardware–software integration and gained interdisciplinary teamwork experience over 1.5 years in a multi-disciplinary project team.

Private Mathematics Tutor **2021 - 2025**

Istanbul, Türkiye

- Delivered one-on-one mathematics tutoring by preparing personalized study plans, exercises and progress tracking for students.

EDUCATION

Bachelor of Electronics Engineering
Gebze Technical University
2022-2027

- Focus: VLSI design, analog/RF and digital integrated circuits
- Relevant courses: Digital Microelectronic Circuits, Simulation of Electronic Circuits, Electronics II

High School
Kadıköy Anatolian Imam Hatip High School
2017-2021

PROJECTS

ML-Based Aging Simulation and Self-Healing Analog Circuits

Graduation Project (In Progress) | Advisor: Prof. Dr. Engin Afacan | Python, ngspice

- Building an open-source aging simulation flow that applies BTI and HCI degradation models to analog circuits such as an OTA, bandgap reference, comparator and ring oscillator.
- Generating labeled datasets with Monte Carlo and Latin hypercube sampling, and comparing machine learning models to predict circuit aging much faster than simulation.
- Designing a closed-loop self-healing controller that compensates aging-induced degradation across different circuit topologies.

EMI Filter Design for a 28 V DC Power Input

Power Electronics / EMC Project | LTspice, KiCad

- Designed a filter to reduce conducted EMI on an avionics-grade DC power input, analyzing differential-mode and common-mode noise and their return paths.
- Compared LC, π, ferrite-bead and common-mode choke topologies, and selected components based on current, voltage and parasitic constraints.
- Simulated frequency and transient response in LTspice and prepared the schematic and PCB layout in KiCad with EMC-aware grounding and routing, referencing MIL-STD-461/704.

2.4 GHz CMOS Low-Noise Amplifier (LNA) Design

RF Integrated Circuit Design Project | 90 nm CMOS

- Designed a cascode LNA with inductive source degeneration for the 2.4 GHz ISM band, from hand calculations to final simulation.
- Optimized input matching, gain, noise figure and linearity while staying well within the power budget.
- Achieved unconditional stability using stabilization resistors and verified linearity with two-tone FFT analysis.

8-Bit ALU Design and Layout

VLSI Project | Synopsys Custom Compiler, StarRC

- Designed an 8-bit ALU with adder/subtractor, multiplier, logic unit, shifter and multiplexer from RTL schematics to transistor level.
- Drew full-custom layouts of the sub-blocks and verified them through DRC, LVS, parasitic extraction and post-layout simulation.
- Evaluated area–delay trade-offs and applied improvements to reduce delay.

Digital Standard-Cell Library in 90 nm CMOS

Full-Custom IC Design Project | Synopsys Generic 90 nm PDK

- Designed, laid out and verified basic logic cells (inverter, NAND, NOR, XOR/XNOR, MUX), meeting all delay specifications after layout.

Transistor-Level LM741 Op-Amp Modeling and Verification

Simulation of Electronic Circuits Final Project | LTspice

- Built the LM741 op-amp at transistor level and verified its key parameters against the manufacturer datasheet.
- Designed an active low-pass filter on the verified model, with simulation results matching theoretical calculations.

Low-Power Active-Loaded Differential Amplifier

Analog Circuit Design Project | LTspice

- Designed a differential amplifier meeting gain, bandwidth and sub-milliwatt power targets, selecting transistors by performance and cost.

Embedded Systems & Robotics Projects

Academic Projects | C, STM32, Arduino

- Developed STM32-based applications for sensor reading, threshold-based alert systems and microcontroller-controlled control tasks.
- Designed a line-follower robot covering sensor data processing, control algorithms and hardware–software integration.

SKILLS

- **IC Design & EDA:** Synopsys Custom Compiler, HSPICE, PrimeWave, StarRC, LTspice, ngspice
- **VLSI & Layout:** Full-custom layout, DRC, LVS, parasitic extraction (PEX), post-layout verification
- **Analog & RF Design:** LNA design, S-parameters, noise figure, op-amp characterization, active filters
- **PCB & EMC Design:** KiCad, schematic and PCB layout, conducted EMI filter design (DM/CM), grounding and layout for EMC
- **Programming & Embedded:** C (Standard/Embedded), Python, ARM Assembly, STM32, Arduino, PyBullet

LEADERSHIP, VOLUNTEERING & ADDITIONAL EXPERIENCE

Robotics and Automation Club — Active Member & Kelebek Robotics Olympics (KELEBEKRO) Referee (May 2025)

European Solidarity Corps — Cultural Heritage Volunteer (Action to Protect)
Galicia, Spain | 2024

- Contributed as part of a team to the protection of world cultural heritage, sustainable rural development and community solidarity, strengthening intercultural communication and teamwork.

IEEE Gebze Technical University Student Branch & Fine Arts Society — Member

LANGUAGES

Turkish: Native **English:** Professional Working Proficiency (YÖKDİL: 88.75/100, 2026) **Arabic:** Beginner